

1. A computer system has the main memory of size 150MB, which has been allotted to three processes P_1 , P_2 and P_3 . As shown in below, each process has been allocated some size of main memory, and the maximum size of main memory needed by each process is also given.

It is assumed that: (1) the main memory having been allocated to a process cannot be preempted by the other processes; (2) swapping in and swapping out of processes are not allowed.

Process	<u>Maximum Memory Needed</u>	<u>Memory Allocated</u>
P_1	70MB	45MB
P_2	60MB	40MB
P_3	60MB	15MB

When another process P_4 enters into the system, it immediately requests main memory of the size 25MB, and its maximum size of main memory needed is 60MB.

Answer the following questions, by means of the banker's algorithm:

- (1) What are the contents of the matrix *Allocation*, *Max*, *Need* and the vector *Available*?
- (2) Can the P_4 's request of main memory be granted immediately? Will be the system in a safe state? and why?

问题描述:

2 假设有三个美食家和一个供应者。每个美食家不停地包饺子并吃饺子，但是要包起一个韭菜鸡蛋饺子，美食家需要三种材料：韭菜、鸡蛋和饺子皮。三个美食家中，第一个拥有韭菜、第二个拥有鸡蛋、第三个拥有饺子皮。供应者无限地提供这三种材料，每次将两种不同的材料放到桌子上，拥有剩下那种材料的美食家将会包一个饺子并吃掉它，并给供应者一个信号告诉完成了，供应者就会继续放两种材料在桌上，让一位美食家包饺子并吃饺子，这种过程一直重复下去。、

采用信号量，给出信号量初值，给出相关进程描述。